Lab 1 Submission

Requirements Engineering & UML Use-Case Modelling — Problem Statement #42

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SECTION : B

System: Internal Microservice Catalog & Health Portal

1. Requirements

ID	Type	Description	Priorit	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ping microservice health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.	High	Pass: health checks occur at the required interval and the rolling 24-hour availability percentage is updated. Fail: an unreachable service is shown as healthy or the availability value is not updated.	This is the functional requirement supplied in the problem statement.
FR-002	Functional	The system shall maintain a catalog of the microservices covered by the developer portal and their dependency relationships.	High	Pass: a registered microservice and its dependency links can be viewed in the portal. Fail: a listed dependency is missing.	The portal is described as a microservice catalog that maps service dependencies.
FR-003	Functional	The system shall display the dependency relationships between the microservices in the developer portal.	High	Pass: the dependency view shows the services and their stated relationships. Fail: a relationship recorded in the catalog is absent from the view.	Dependency mapping is one of the main functions stated in the problem context.
FR-004	Functional	The system shall aggregate API documentation for the microservices in the developer portal.	High	Pass: API documentation associated with a microservice can be accessed from the portal. Fail: the associated documentation is not available from the portal.	The problem context explicitly requires API documentation aggregation.
FR-005	Functional	The system shall dispatch a downtime alert when a microservice has three consecutive failed health probes.	High	Pass: after the third consecutive failed probe, downtime is recorded and an alert is dispatched. Fail: an unreachable service is marked healthy or the alert is not dispatched after the third consecutive failure.	The three-failure alert condition is explicitly given in the problem statement.
NFR-001	Nonfunctional – Performance	The service catalog dependency graph viewer shall support interactive node architectures containing up to 200 services.	High	Pass: benchmarking confirms that a 200-service graph remains interactive under the agreed test conditions. Fail: the graph becomes unusable before 200 services.	This is the non-functional requirement supplied in the problem statement.
NFR-002	Nonfunctional – Timeliness	After the third consecutive failed health probe is detected, the downtime alert shall be dispatched within the same health-monitoring cycle.	High	Pass: the alert is dispatched before that monitoring cycle completes. Fail: the third failure is recorded but the alert is deferred to a later cycle.	The timing requirement is derived only from the stated automated health-check and downtime-alert behaviour; no unrelated system feature is added.

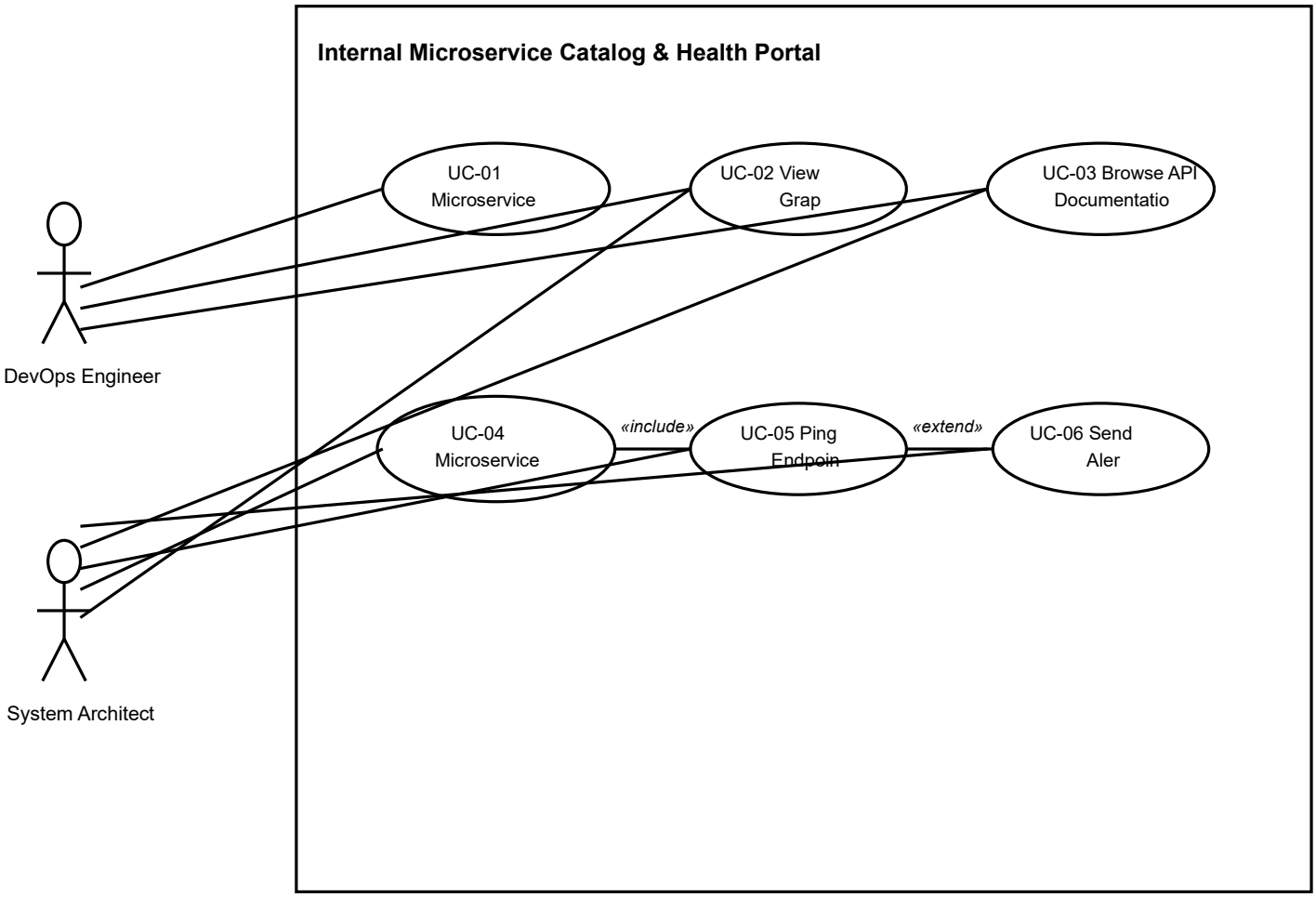
2. UML Model

Actors: DevOps Engineer; System Architect.

Core use cases: Maintain Microservice Catalog; View Dependency Graph; Browse API Documentation; Monitor Microservice Health; Ping Health Endpoint; Send Downtime Alert.

Relationships: UC-04 «include» UC-05; UC-06 «extend» UC-04.

UML Use-Case Diagram — Internal Microservice Catalog & Health Portal



Solid = association Dashed = «include» / «extend»

3. Use-Case Flow — UC-04 Monitor Microservice Health

Primary Actor: DevOps Engineer

Goal: Periodically check microservice health, record availability, and trigger the stated downtime alert condition.

Preconditions

- The microservice has a health endpoint.
- The health-monitoring process is running.

Postconditions

- The health-check result is recorded.
- The rolling 24-hour availability percentage is updated.
- Three consecutive failures cause downtime to be recorded and an alert to be dispatched.

Main Success Scenario

1. The health-monitoring process starts the scheduled check.
2. The system pings the microservice health endpoint.
3. The endpoint responds successfully.
4. The system records the successful health result.
5. The rolling 24-hour availability percentage is recalculated.
6. The service continues to be treated as healthy until a later check reports otherwise.

Alternate Flow

1. The health endpoint cannot be reached or does not return a successful response.
2. The system records the failed probe.
3. The system waits for the next scheduled health check.
4. The same microservice fails the second and then the third consecutive probe.
5. The system records the downtime and dispatches the downtime alert.

Note: No additional user roles, search functions, service history, authentication requirements, or other features have been added to this final version.